

ANX-PR/CL/001-01

LEARNING GUIDE

SUBJECT

83000117 - Integrated Management Of Marine Renewable Installations

DEGREE PROGRAMME

08IN - Master Universitario En Ingenieria Naval Y Oceanica

ACADEMIC YEAR & SEMESTER

2025/26 - Semester 1

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1. Description

1.1. Subject details

Name of the subject	83000117 - Integrated Management Of Marine Renewable Installations
No of credits	6 ECTS
Type	Optional/elective
Academic year of the programme	Second year
Semester of tuition	Semester 3
Tuition period	September-January
Tuition languages	English
Degree programme	08IN - Master Universitario en Ingeniería Naval y Oceanica
Centre	08 - E.T.S. De Ingenieros Navales
Academic year	2025-26

2. Faculty

2.1. Faculty members with subject teaching role

Name and surname	Office/Room	Email	Tutoring hours *
Jaime Moreu Gamazo		jaime.moreu@upm.es	Sin horario.
Julio Garcia Espinosa (Subject coordinator)	PP 82	julio.garcia.espinosa@upm.es	Sin horario. Según horarios
Jose Luis Moran Gonzalez		joseluis.moran@upm.es	Sin horario.

* The tutoring schedule is indicative and subject to possible changes. Please check tutoring times with the faculty member in charge.

3. Skills and learning outcomes *

3.1. Skills to be learned

CTUPM12 - Gestión económica y administrativa. Los estudiantes son capaces de realizar las operaciones administrativas vinculadas a la gestión económica y de recursos humanos. Economic and administrative management. Students are able to perform administrative operations related to economic and human resources management.

CTUPM13 - Trabajo en contextos internacionales. Los estudiantes son capaces de integrarse en un grupo o equipo, colaborando y cooperando con otros. Tienen la capacidad para trabajar con estudiantes de otras disciplinas y de aceptar la diversidad social y cultural. Work in international contexts. Students are able to integrate into a group or team, collaborating and cooperating with others. They have the ability to work with students from other disciplines and to accept social and cultural diversity.

3.2. Learning outcomes

RA59 - S7 Comunicación oral y escrita.

RA31 - C: Capacidad para resolver problemas complejos y para tomar decisiones con responsabilidad sobre la base de los conocimientos científicos y tecnológicos adquiridos en materias básicas y tecnológicas aplicables en la ingeniería naval y oceánica, y en métodos de gestión

* The Learning Guides should reflect the Skills and Learning Outcomes in the same way as indicated in the Degree Verification Memory. For this reason, they have not been translated into English and appear in Spanish.

4. Brief description of the subject and syllabus

4.1. Brief description of the subject

This course reviews the main aspects of the planning, development, and management of a marine renewable power generation park. It examines the primary objectives and tasks for assessing its technical and financial feasibility. Additionally, it presents the various development phases of a project of this nature.

Esta asignatura revisa los aspectos principales del proyecto, desarrollo y gestión de un parque de dispositivos de generación renovable marina. Para ello, analiza cuáles son los objetivos y tareas principales para el estudio de su viabilidad técnica y financiera. Asimismo, presenta las diferentes fases de desarrollo de un proyecto de esta naturaleza.

4.2. Syllabus

1. Introduction

2. Project development

2.1. Development phase. Activity outline: pre-development (feasibility and permitting) and pre-construction (detailed project and financial closing).

2.2. Execution Phase / Operation Phase. Engineering and design, construction, installation, commissioning, grid connection, start of commercialization. Activity outline, principles of predictive and corrective maintenance. Key milestones: FID, COD.

3. Technical feasibility

3.1. Review of available technologies. Characteristics, main components, power and performance curves, utilization thresholds, production estimation (capacity factor).

3.2. Types of substructures. Fixed and floating, types, characteristics, manufacturing, and installation (e.g., types of piles, gravity-based, anchors, etc.), example..

3.3. Site selection. Resource estimation, calculation and energy density maps, GIS tools and resource assessment. Other site characteristics.

3.4. Electrical interconnection of the park. Grid layout (array configuration), components, transmission capacity, and performance.

3.5. Energy conversion and grid connection. Interconnections between parks, electrical substations, converter stations, equipment sizing, grid connection and feed-in (surpluses). H2 generation.

3.6. Review and discussion of case studies.

4. Permitting

4.1. Entorno regulatorio. POEM, modelos regulatorios en Europa, regulación fiscal, contratos de exclusividad de uso marítimo, regimen de permisos y entidades responsables (España), interacción con el entorno social.

4.2. Regulatory environment. POEM, regulatory models in Europe, tax regulation, maritime exclusive use contracts, permit regime and responsible entities (Spain), interaction with the social environment.

5. Financial feasibility

5.1. Cost assessment. Budget items (DEVEX, CAPEX, OPEX, DECEX), principles of economic evaluation of a project, concept of IRR, NPV, WACC.

5.2. Financing and refinancing. Understanding the ways to finance a project (corporate, project-linked bonds, internal financing, non-recourse financing, limited recourse financing, project finance structure, etc.). Grants.

5.3. Corporate structures and business partnerships. SPV, joint venture, business alliances, objectives, and operations.

5.4. Contracts and other aspects. Contracts in development phase, execution, O&M, EPC, multi-contract strategies, power purchase agreements (PPA).

5.5. Review and discussion of case studies.

5. Schedule

5.1. Subject schedule*

Week	Type 1 activities	Type 2 activities	Distant / On-line	Assessment activities
1	Tema 1 Duration: 04:00 Lecture			
2	Tema 2 Duration: 04:00 Lecture			
3	Tema 3. Revisión de las tecnologías disponibles Duration: 04:00 Lecture			
4	Tema 3. Tipologías de subestructuras Duration: 04:00 Lecture			
5	Tema 3. Selección del emplazamiento Duration: 04:00 Lecture			
6	Tema 3. Interconexión eléctrica del parque. Conversión energética y conexión a red. Duration: 04:00 Lecture			
7	Tema 3. Conversión energética y conexión a red y Revisión y discusión de casos de estudio Duration: 04:00 Cooperative activities			
8	Tema 3. Revisión y discusión de casos de estudio Duration: 04:00 Cooperative activities			
9	Tema 4. Entorno regulatorio Duration: 04:00 Lecture			
10	Tema 4. Modelos de subastas y otros aspectos Duration: 02:00 Lecture Tema 5. Evaluación de costes Duration: 02:00 Lecture			

11	Tema 5. Evaluación de costes. Financiación y refinanciación. Duration: 04:00 Lecture			
12	Tema 5. Estructuras societarias y alianzas comerciales. Contratos y otros aspectos. Duration: 04:00 Lecture			
13	Revisión y discusión de casos de estudio Duration: 04:00 Cooperative activities			
14	Presentación e informe del trabajo individual Duration: 04:00 Additional activities			Presentación del trabajo individual Individual presentation Progressive assessment Presential Duration: 04:00 Informe del trabajo individual Individual work Progressive assessment Presential Duration: 00:00
15	Seminario Duration: 02:00 Lecture			
16	Examen individual Duration: 02:00 Additional activities			Examen individual Written test Progressive assessment Presential Duration: 02:00
17	Examen individual Duration: 02:30 Additional activities			Examen individual Written test Global examination Presential Duration: 02:30

Depending on the programme study plan, total values will be calculated according to the ECTS credit unit as 26/27 hours of student face-to-face contact and independent study time.

6. Activities and assessment criteria

6.1. Assessment activities

6.1.1. Assessment

Week	Description	Modality	Type	Duration	Weight	Minimum grade	Evaluated skills
14	Presentación del trabajo individual	Individual presentation	Face-to-face	04:00	30%	/ 10	CTUPM12 CTUPM13
14	Informe del trabajo individual	Individual work	Face-to-face	00:00	35%	/ 10	CTUPM12 CTUPM13
16	Examen individual	Written test	Face-to-face	02:00	35%	/ 10	CTUPM12 CTUPM13

6.1.2. Global examination

Week	Description	Modality	Type	Duration	Weight	Minimum grade	Evaluated skills
17	Examen individual	Written test	Face-to-face	02:30	100%	5 / 10	CTUPM12 CTUPM13

6.1.3. Referred (re-sit) examination

Description	Modality	Type	Duration	Weight	Minimum grade	Evaluated skills
Examen individual	Written test	Face-to-face	02:30	100%	5 / 10	CTUPM12 CTUPM13

6.2. Assessment criteria

Progressive evaluation

The progressive evaluation of the course will consist of two exams:

- Completion and presentation of an individual project that applies the knowledge acquired in the course. Evaluation will be based on both the written report of the project (weighting 35%) and a brief oral presentation (weighting 30%).
- A written exam covering various topics discussed in the course, with a weighting of 35% towards the final grade.

To pass the course, it will be necessary to achieve a weighted average grade of at least 5.0, according to the specified weightings.

Global and extraordinary assessment

The global and extraordinary assessment will be conducted through a written exam covering various topics discussed in the course, where a minimum grade of 5.0 must be attained.

Evaluación progresiva

La evaluación progresiva de la asignatura se realizará mediante dos pruebas:

- La realización y presentación de un trabajo individual que ponga en práctica los conocimientos adquiridos en la asignatura. Se evaluará tanto el informe del trabajo realizado (peso del 35%) como una presentación oral breve (peso del 30%).
- * Un examen escrito sobre los diferentes temas tratados en la asignatura, con un peso en la nota del 35%.

Para aprobar la asignatura será necesario obtener una nota media ponderada, de acuerdo a los pesos indicados, de al menos 5.0.

Evaluación global y extraordinaria

La evaluación global y extraordinaria se realizará mediante un examen escrito sobre los diferentes temas tratados

en la asignatura, en el que habrá que obtener una nota de al menos 5.0.

7. Other information

7.1. Other information about the subject

The review of the different course contents will be carried out through conferences or seminars specifically focused on each topic.

La revisión de algunos contenidos de la asignatura se llevará a cabo a través de conferencias o seminarios enfocados específicamente en cada temática.